

PARVESH SHANMUGAM

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EDUCATION

B.Tech – CSE (Cyber Physical Systems), VIT Chennai | CGPA: 8.20

2023 – 2027

Class XII – Velammal Vidhyalaya Annexure, Chennai | **90.4%**

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, TypeScript

Databases: MySQL, PostgreSQL, SQLite, MongoDB

Web & Frameworks: React, Next.js, Node.js, Express.js, Django, FastAPI, Spring Boot, MERN Stack

Tools: Git, GitHub, Linux, Postman, Docker, VS Code, IntelliJ

AI/ML & Generative AI : Machine Learning, Deep Learning, CNN, LSTM, GNN, YOLO, Federated Learning, XAI, LLMs, RAG

Security: Blockchain, Post-Quantum Cryptography, IoT Security, Edge AI, AI-based IDS

RESEARCH & PUBLICATIONS

Secure Smart City Data-Sharing Platform

Flask, REST APIs | Book Chapter 2025

- Privacy-preserving data exchange reducing integration time by 30% and latency by 25%; co-authored CPS book chapter on Critical Infrastructure Protection.

FedUnlearnGuard – Byzantine-Resilient Federated Unlearning

PyTorch, Docker, Hyperledger

- GDPR-compliant federated learning framework with 4-layer defense; reduced poisoning attacks from 92% to 3% with minimal accuracy loss.

Physics-Aware GPS Spoofing Detection for UAVs

ML, Explainable AI

- Interpretable trajectory-physics framework achieving ~95% spoofing detection via motion-constraint feature engineering and real-time streaming.

Automated OCT Skin Layer Segmentation for Diabetic Foot Analysis Imaging | Journal Paper Under Review

Deep Learning, U-Net Architectures, Biomedical

- Developed a deep learning pipeline using U-Net-based architectures for automated segmentation of OCT skin layers and quantitative analysis of microstructural changes associated with diabetic foot.

Fairness-Aware Quantum-Inspired Optimization for Smart City Digital Twins

QIGA, OpenDSS

- Multi-objective quantum-inspired framework for equitable infrastructure resource allocation across smart city systems.

Quantum-Resistant Blockchain Framework for IoT-Edge Systems

Post-Quantum Crypto, AI IDS

- Hardware-software co-design: lattice-based cryptography + blockchain authentication + AI IDS + vision-based monitoring for secure autonomous systems.

WORK EXPERIENCE

Full-Stack Developer Intern – IXES Global Solutions

Sep – Nov 2025 | Hybrid

- Built production full-stack applications; handled database design, REST APIs, and dynamic UI with focus on scalability and performance.

Full-Stack Developer Intern – ennVee Solutions Pvt. Ltd.

May – Jun 2025 | Offline

- Delivered Skills Matrix Portal (Django/MySQL) and role-based Task Management System (React/FastAPI/MySQL); analyzed Viva Learning adoption via Jira & Rally.

Research & Development Intern – VIT Chennai (Smart City)

May – Jul 2024 | Virtual

- Built Flask REST API platform cutting integration time by 30% and latency by 25%; co-authored Cyber Physical Systems academic book chapter.

Research Assistant – VIT Chennai (Medical Imaging & ML)

Mar 2025 – Present | Virtual

- Annotating OCT and thermal foot images; training ML models to detect diabetic foot ulcers and support clinical severity assessment.

PROJECTS

MR GLOSS (Next.js, Three.js, SEO) – High-performance business site with 3D animations, chatbot integration, and SEO optimization (<https://www.mrglossceramiccoating.com/>).

MERCARA EXPORTS (React, Tailwind, Full Stack) – Scalable B2B export platform with product catalogs and premium UI for agro-commodity operations.

Dental Clinic Management – Dr. Bharads (Full Stack) – Production app for appointment handling, service management, and clinic digital presence (<https://www.smileywood.in/>).

ZYPH26 Summit App (Full Stack) – Official event application powering the Nilgiris Calls Global Summit at KCT Tech Park, Coimbatore (<https://zyph26.texus.io/>).

AI-Driven Nursing Competency Evaluation and Adaptive Training System (LLMs) – AI-powered platform that assesses nurses with adaptive quizzes and automatically generates personalized learning modules to improve their clinical competencies.

ACHIEVEMENTS

- Co-authored book chapter: "Critical Infrastructure Protection: CPS Stability, Security, Privacy & Forensics" – publication 2025.
- Best Paper Award – "AI-Powered Traffic Signals and Street Lights" at International SCES 2024 (Feb 2024).
- Proud Sponsor, Nilgiris Calls Global Summit | ZYPH Hackathon, KCT Tech Park – built and delivered zyph26.texus.io.